# Potentially new constructions

Low rank Grothendieck rings · Novelty shortlist and unresolved cases · 19 October 2026

**To the best of our current knowledge, these seven constructions are promising candidates for new tensor categories.** Their novelty up to tensor equivalence remains subject to independent verification.

## Reported constructions to examine for novelty

m denotes maximum fusion multiplicity. Click a census ID for its construction source.

| | |
|---|---|
| **R6_4_249**<br>Rank 6 · m = 4 | **Sign-swap equivariantization** of the diagonal-fermion quotient of two identical half-$E_6$ categories. Known inputs: Aasen-Lake-Walker. |
| **R6_4_322**<br>Rank 6 · m = 4 | **Mixed diagonal-fermion quotient** of Ad($E_6$) and PSU(2)$_6$. Both input categories are already known. |
| **R6_6_1712**<br>Rank 6 · m = 6 | **Relative Fibonacci center of $TT_3$.** The input $TT_3$ is the known dual even part of 2D2; see Edie-Michell, Definition 2.5. |
| **R7_4_35**<br>Rank 7 · m = 4 | **Inversion descent and reflection Morita dual** of an orbifold of NG($C_4 \times C_4$,16). Input: Yu-Zhang. |
| **R7_4_2665**<br>Rank 7 · m = 4 | **Analogous inversion/Morita construction** from NG($C_{16}$,16), with different invertibles and duality. Input: Budinski (2021). |
| **R8_2_37**<br>Rank 8 · m = 2 | **Nonsplit $C_2$-extension of 2D2,** with an odd invertible a satisfying $a^2 = g$. Uses Grossman-Izumi-Snyder and Grossman. |
| **R8_2_575**<br>Rank 8 · m = 2 | **Explicit Davydov-Runkel extension** of Kashina's self-dual 16-dimensional Hopf algebra. Simple dimensions: (1,1,1,1,2,2,2,4). |

## Known model added in the rank-seven extension

**R7_5_5678 · Rank 7 · m = 5.** Rep(AGL(1,7)) = Rep($C_7 \rtimes C_6$), of global dimension 42. The full fusion tensor is verified. This is a known representation category, not an additional novelty candidate.

## OPEN: 17 fusion rings with no established categorification

**R6_4_155 · Rank 6 · multiplicity 4.** Generalized near-group ring R′(4): four invertibles forming $C_2 \times C_2$, and two mutually dual noninvertibles of dimension $4 + 3\sqrt{2}$. The ring already appears in Meng-Yu, Theorem 1.1.

**R6_8_1660 · Rank 6 · multiplicity 8.** Self-dual ring containing an Ising fusion subring. A further simple X has dimension $3 + 2\sqrt{2} + 2\sqrt{3} + \sqrt{6}$ and satisfies $X^2 = 1 + [3(1 + g) + 4\sigma]X$, where $\sigma^2 = 1 + g$.

**Rank 7 · multiplicity 5: 15 unresolved rings.** Of all 5,805 new rings, 5,789 are excluded and one is realized. The remaining census IDs are:

R7_5_443 · R7_5_1214 · R7_5_1964 · R7_5_3895 · R7_5_4530
R7_5_4531 · R7_5_4569 · R7_5_4653 · R7_5_4982 · R7_5_5029
R7_5_5039 · R7_5_5040 · R7_5_5343 · R7_5_5623 · R7_5_5778

All 17 remain open for general complex categorification; no positive model or complete exclusion is established. Exact fusion data.

**AI-generated research · independent verification pending.**
Independent human verification and Lean formalization remain pending. Please treat the claims with caution.